

The empirical recurrence for OEIS A232347: recovering a conjecture that is not written down

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Abstract

OEIS A232347 records “Empirical recurrence of order 80”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 102$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 80, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 80 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A232347 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with every element both $\geq$ and $\leq$ some horizontal, vertical, diagonal or antidiagonal neighbor.”. It has offset 1 and begins

1461, 111607, 6753291, 426813751, 27382170933, 1753672323629, 112198453386465, 7178913247853751, ...

The entry states:

Empirical recurrence of order 80 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:20:19, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 6643$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 102$ of the 6643, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 102$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 13286$ exact terms of the model returns order 80 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	53	c_{21}	−8452885010280586	c_{41}	20043050476463538229	c_{61}	−208008113932962
c_2	443	c_{22}	−22441164987095805	c_{42}	14829521201865601175	c_{62}	383869438466539
c_3	13520	c_{23}	37366212642765745	c_{43}	−1116513657421769045	c_{63}	−77913685531634
c_4	196960	c_{24}	44828609805800166	c_{44}	−69338210429104472478	c_{64}	−12230044299654
c_5	287927	c_{25}	58640857741794638	c_{45}	−21270488836044414196	c_{65}	148650904173466
c_6	−5774392	c_{26}	324961748021295931	c_{46}	−13265506539197129333	c_{66}	−24126237915772
c_7	−45080988	c_{27}	−406818511558730193	c_{47}	−8567657089741381565	c_{67}	−12248396237464
c_8	−583866881	c_{28}	−1080392457450794041	c_{48}	35400028879245297507	c_{68}	99323804171635
c_9	−2236415936	c_{29}	−275959985551463081	c_{49}	11603053777387533019	c_{69}	−4427056817955
c_{10}	12898082481	c_{30}	−1955097201431239784	c_{50}	3406397480560675627	c_{70}	7792010026624
c_{11}	66350011478	c_{31}	2289366434044909184	c_{51}	6716495366910166172	c_{71}	9791804093132
c_{12}	195289442627	c_{32}	8789020465020093673	c_{52}	−8870725045467898355	c_{72}	−4453069985280
c_{13}	1687414150787	c_{33}	1451791357412831429	c_{53}	−5357297570337305289	c_{73}	−273584268697
c_{14}	−4045815254794	c_{34}	5865949750456208724	c_{54}	566574985406627960	c_{74}	1727063166976
c_{15}	−57903674762723	c_{35}	−6698525486540762818	c_{55}	−2073644983578907001	c_{75}	−155513873203
c_{16}	−8303911140525	c_{36}	−34422920353473657626	c_{56}	785328135484136856	c_{76}	708033363968
c_{17}	529369791589921	c_{37}	−7274073650207036346	c_{57}	1495197444970931021	c_{77}	159455576064
c_{18}	280582797063873	c_{38}	−10208452904373016659	c_{58}	−291732279243603328	c_{78}	−64828538880
c_{19}	−1124601142382730	c_{39}	8927184105387154548	c_{59}	257324317460960715	c_{79}	−8650358784
c_{20}	1033919979541646	c_{40}	69120690843411374818	c_{60}	39254307765981977	c_{80}	2477260800

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 80, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $80 + 4$ terms removed returns order 80 as well, so $\deg q_{\min} = 80 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $80 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 80 that this sequence satisfies — and that family turns out to have exactly one member.

References

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